Running FV3-JEDI 3D-Var-Hyb:
Hands-on sessions with C96C48_ufs_hybatmDA

https://github.com/NOAA-EPIC/CADRE-DA-training

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Outline

- Six Experiment Sets at a Glance — Overview of all experiments and YAML changes
  - Control experiment configuration: README, additional single obs experiment is available
- Diagnostics-Driven Sessions with UFS-DA Diagnostics toolkit — JEDI log parser, plotting scripts, OMB/OMA stats, etc.
  - Power Spectra Analysis — Quantifying scale distribution across experiments
  - χ^2 Consistency & Jo/P Diagnostics — Observation error calibration and Talagrand's criterion (2003, Objective Validation and Evaluation of Data Assimilation)
  - Obs-Space Diagnostics: histograms, extended RMS stats, ATMS scan-position check, lat-binned stats
  - Together with increment plots and zonal-mean diagnostics
- Day 1: System Access and Control Run — C96C48 Hybrid 3D-Var baseline, YAML overview, Diagnostics - increment maps, JEDI log walkthrough, etc.
- Day 2: B Error Experiments — SABER/BUMP NICAS Horizontal length scale & hybrid weight tuning
- Day 3: Obs Experiments — ATMS thinning, obs error scaling, obs-type toggle
- Take home messages

Six Experiment Sets at a Glance

Experiment Sets	What changes (YAML)	Diagnostic Focus
Control C96/C48 Hybrid 3DVar	Baseline — no changes	Increment maps, OMB/OMA stats, chi-squared consistency, power spectra (baseline reference)
Horizontal Length Scale	`horizontal_length_scale` in NICAS block	Increment smoothness, Jo/Jp ratio under broadened/narrowed length scale, power spectra (scale shift)
Hybrid Weight	`ensemble_weight` / `static_weight` ratio	Increment magnitude, ensemble vs static contribution, power spectra (static vs ensemble structure)
ATMS Thinning	`obs thinning` parameters	Observation density maps, analysis sensitivity, power spectra (small-scale roll-off)
ATMS Obs Error	`obs error.original` (scaling factor)	OMB/OMA histograms, chi-squared consistency, power spectra (high-wavenumber energy)
Obs-Type Toggle (Optional)	`obs types` list (add/remove sensors)	Observation count impact, coverage maps, power spectra (variable-specific constraint)

Power Spectra Analysis – Quantification of Scale Distribution

2-D Fourier Transform

For a horizontal field $f(x,y)$ on an $N_x \times N_y$ grid, the 2-D Discrete Fourier Transform is

$$F(k_x, k_y) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} f(x,y)\, e^{-i2\pi\left(\frac{k_x x}{N_x} + \frac{k_y y}{N_y}\right)}$$

Each pair (k_x, k_y) corresponds to one *grid-scale sinusoidal wave* with a specific wavelength and direction.

Power Spectrum

The 2-D power spectrum is

$$P(k_x, k_y) = |F(k_x, k_y)|^2$$

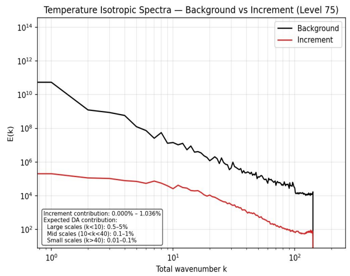
This represents the energy contribution of each grid-scale wave.

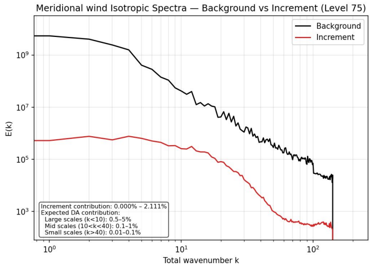
Isotropic Spectrum

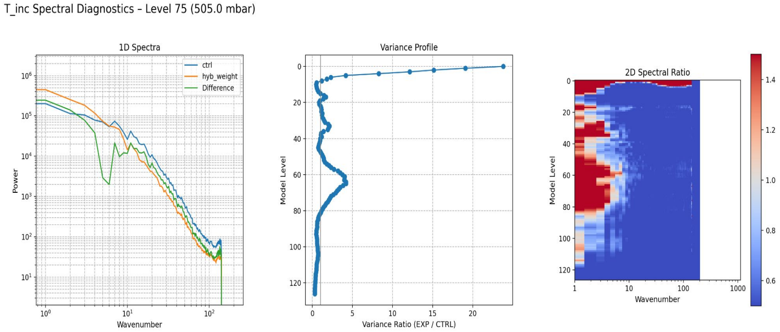
The 2-D spectrum is radially averaged into bins of total wavenumber

$$K = \sqrt{k_x^2 + k_y^2}$$

to produce a 1-D isotropic spectrum $P(K)$. This gives the distribution of variance across spatial scales.







See: https://ufs-da-diagnostics.readthedocs.io/en/latest/diagnostics_overview.html#overview

Power Spectra Analysis — Quantification of Scale Distribution

Experiment Sets	Spectral Effect	What participants should see
Control C96/C48 Hybrid 3DVar	Baseline spectral signature	Reference power spectrum — establishes expected energy distribution across wavenumbers
Horizontal Length Scale	**B** correlation cutoff shifts	Most dramatic shift — doubling length scale suppresses small-scale energy, shifts power to larger scales; demonstration of **B** as spatial filter
Hybrid Weight	Static vs. ensemble **B** ratio	More static weight → smoother spectrum (large-scale); more ensemble weight → more small-scale flow-dependent structure
ATMS Thinning	Observation density at small scales	Fewer obs → less small-scale constraint → spectrum rolls off earlier
ATMS Obs Error	Effective obs weight across scales	Halving **R** → more small-scale structure from observations → spectrum gains energy at higher wavenumbers
Obs-Type Toggle (Optional)	Sensor-specific spectral constraint removed	Removing an obs type removes spectral constraint in its sensitive region

Key Insight: Power spectra make the **B** vs. **R** balance visible at every scale — participants can immediately see where the analysis is driven by background (large scales) versus observations (small scales).

Chi-Squared Consistency & J_o/p Diagnostics

Chi-Square Consistency Check

The chi-square consistency diagnostic evaluates whether the innovations (OMB) are broadly compatible with the assumed observation-error variances. It provides a quick indication of whether the specified observation-error model is reasonable.

For each observation i, the innovation d_b is scaled by its assumed observation-error variance σ_{o,i}². The normalized innovation is

$$z_i = \frac{d_{b,i}}{\sigma_{o,i}}.$$

If the observation-error model is reasonable, the average value of z_i² should be close to one. This leads to the chi-square consistency measure

$$\chi^2 = \frac{1}{N}\sum_{i=1}^{N} \frac{d_{b,i}^2}{\sigma_{o,i}^2}.$$

Interpretation

- $\chi^2 \approx 1$ → observation-error variances are broadly consistent with the size of the innovations.
- $\chi^2 >> 1$ → observation-error variances are likely too small (innovations too large).
- $\chi^2 << 1$ → observation-error variances are likely too large (innovations too small).

```
--------------------------------------------------------------------------------
 5. CHI-SQUARED CONSISTENCY (Final Iteration)
--------------------------------------------------------------------------------

 Chi^2 ~ (H(x)-y)^T R^{-1} (H(x)-y).  E[Chi^2/P] = 1.0 if R,B correct.
 JEDI prints Jo = 1/2 * Chi^2, so Chi^2/P = 2 * (Jo/P).
```

Obs Type	Jo	P	Chi2/P	Status
conventional_ps	20377.90	107,491	0.379156	R severely overestimated (too large)
gnssro_cosmic2	60004.30	227,627	0.527216	R overestimated (too large)
ATMS N20	21126.90	167,540	0.252200	R severely overestimated (too large)
ascatw_ascat_metop-b	11358.40	25,650	0.885646	OK (calibrated)
satwind_goes-16	6357.03	191,396	0.066428	R severely overestimated (too large)
TOTAL	119224.53	719,704	0.331315	R severely overestimated (too large)

Chi-Squared Consistency & J_o/p Diagnostics

At convergence, Jo follows a chi-squared distribution with p degrees of freedom — Jo/p ≈ 1 signals correctly specified B and R.: Talagrand (2003) — "Objective Validation and Evaluation of Data Assimilation"

J_o/p Value	Interpretation	Likely Cause	Role of Representativeness Error
≈ 1	**Error covariances consistent** — system well-tuned	B and R correctly specified	Representativeness error is properly included in R for the model resolution
≫ 1	**Observations over-weighted** — system fits obs too strongly	R too small, or B too large (length scales too broad, ensemble weight too high)	Representativeness error underestimated → R too optimistic
≪ 1	**Observations under-weighted** — system trusts background too much	R too large, or B too small (length scales too short, static weight too high)	Representativeness error overestimated or inherited from high-res ops

- How It Maps to FV3-JEDI Experiments:
  - Horizontal Length Scale: Increasing length scale broadens B → Jo/p should decrease, but Jo/p can still increase if the increment becomes less aligned with the observations
  - Hybrid Weight: Shifting toward more static weight changes effective B → watch Jo/p shift
  - Obs-Error Scaling: Reducing ATMS obs errors increases their influence
  - Obs Thinning: Reducing p changes the denominator and obs correlation structure
- Practical Check for Participants:
  - Read Jo and observation count from minimization log
  - Compare Jo/p shifts across experiments
  - Use OMB / OMA histograms as visual proxy — well-tuned system shows OMA variance < OMB variance

Observation-Space Diagnostics

- Histogram Diagnostics
  - OMB/OMA distribution shape, bias, spread
  - Detects gross QC issues, outliers, non-Gaussian behavior
  - Channel-by-channel sanity check for radiances and scalar obs
- Extended RMS Statistics
  - RMS, NRMS, BC-RMS, RMS-diff
  - Highlights systematic biases and error-model mismatches
  - Useful for comparing experiments (e.g., R-tuning, thinning, NICAS scale)
- Scan-Position Diagnostics
  - OMB/OMA vs scan angle
  - Detects cross-track striping, calibration drift, limb effects
  - Essential for ATMS and other cross-track radiometers
- Latitude-Binned Statistics
  - Mean/Std/RMS vs latitude
  - Checks hemispheric asymmetry and regional imbalance
  - Reveals sampling gaps, QC differences, or background error issues
- Together with Increment Diagnostics:
  - Increment maps → spatial structure, flow-dependent features
  - Zonal-mean increments → vertical/latitudinal balance
  - Obs-space stats → confirm whether increments are physically justified

Diagnostics-Driven Sessions & Prepared Tooling

Session	Focus	Diagnostic Outputs
Day 1 — System Access & Control Run	Run baseline C96 hybrid 3DVar; inspect YAML, outputs	Increment maps, OMB/OMA histograms, cost-function convergence
Day 2 — Background Error Modeling	Horizontal Length Scale and hybrid static/ensemble weight experiments	Power spectra, increment spread comparison, jedi log
Day 3 — Observation Experiments	ATMS thinning, obs-error inflation, obs-type toggle	Observation density maps, OMB/OMA shift analysis, power spectra

- Prepared Experiment Scripts and Diagnostic Tooling
  - Slurm sbatch job card driver for all experiments: https://github.com/NOAA-EPIC/CADRE-DA-training/tree/main/year2_cases
  - Experiment running instructions: https://ufs-da-diagnostics.readthedocs.io/en/latest/cadre2026_epic.html
  - Experiment results for reference: https://github.com/NOAA-EPIC/CADRE-DA-training/wiki
  - Synoptic scale weather structures of the control experiment case: https://github.com/jkbk2004/CADRE-DA-training/tree/diag-results/diagnostic/results/year2_cases/day1_ctrl/weather_synoptic_bkg

Day 1: C96/C48 Hybrid 3D-Var Control

Learning Objectives

- Understand hybrid 3D-Var YAML structure
- Run control case end-to-end
- Verify increment fields and Jo convergence
- Record baseline power spectra and Jo/p
- Compute and interpret OMB/OMA statistics

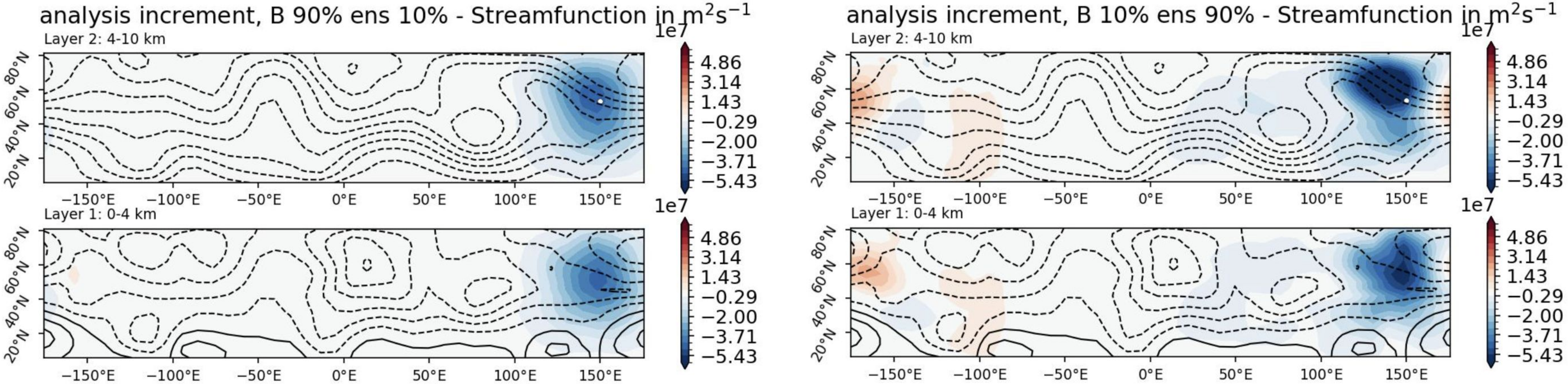
Hands-On Workflow

1. Log in to MSU/Hercules HPC system
2. Review job card, experiment directory structure, and diagnostic tools
3. Inspect YAML configuration
4. Run C96/C48 control case
5. Visualize increments on cubed-sphere grid
6. Generate baseline power spectra
7. Record Jo/p for chi-squared reference
8. Compute OMB/OMA stats as diagnostic check

Shared setup: C96/C48, 80-member ensemble, GFS observations, single cycle

Recall: Assimilating streamfunction observations into two-layer QG model

- Recall what you learned about 3DEnVar in the CADRE-JEDI-Edu tutorial



- Now, we will use a real NWP model (GFS at reduced C96/C48 resolution) and real satellite observations (e.g., ATMS, GNSS-RO).

Comparison to simple model

Similarities:

- Localization length scale and weight to static/ensemble components of B are still important parameters
- YAML files contain similar options (although some differences can be found in the newer JEDI OOPS version used in the following tutorial)

Differences:

- Size of model (e.g., more height levels and variables)
- Not all prognostic variables are observed, some observations (e.g., ATMS) are nonlinear functions of prognostic variables
- Additional parameters related to observation thinning and observation error estimation may require tuning

Interactive Session: Day 1

https://ufs-da-diagnostics.readthedocs.io/en/latest/cadre2026_epic.html

Day1: Login to Hercules (MacBook)

- Open the **Terminal** window

- Login to Hercules (longin nodes) via **SSH**

```
MacBook-Pro ~ % ssh -Y chjeon@hercules-login.hpc.msstate.edu
```

(or) -X

user name

Note: Application for GUI on Mac (optional): XQuartz (https://www.xquartz.org/)

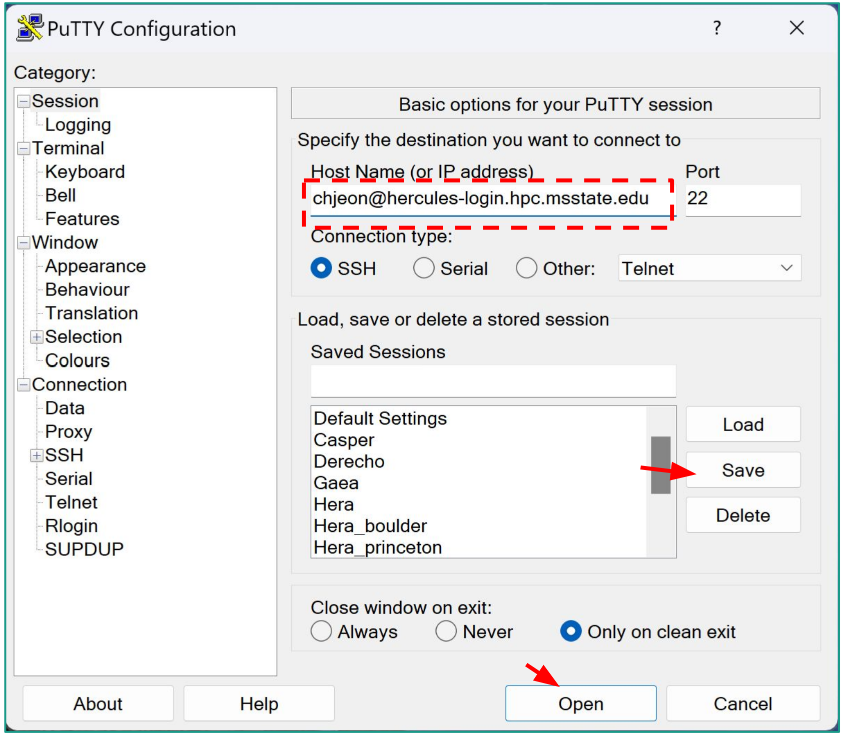
Quick Download

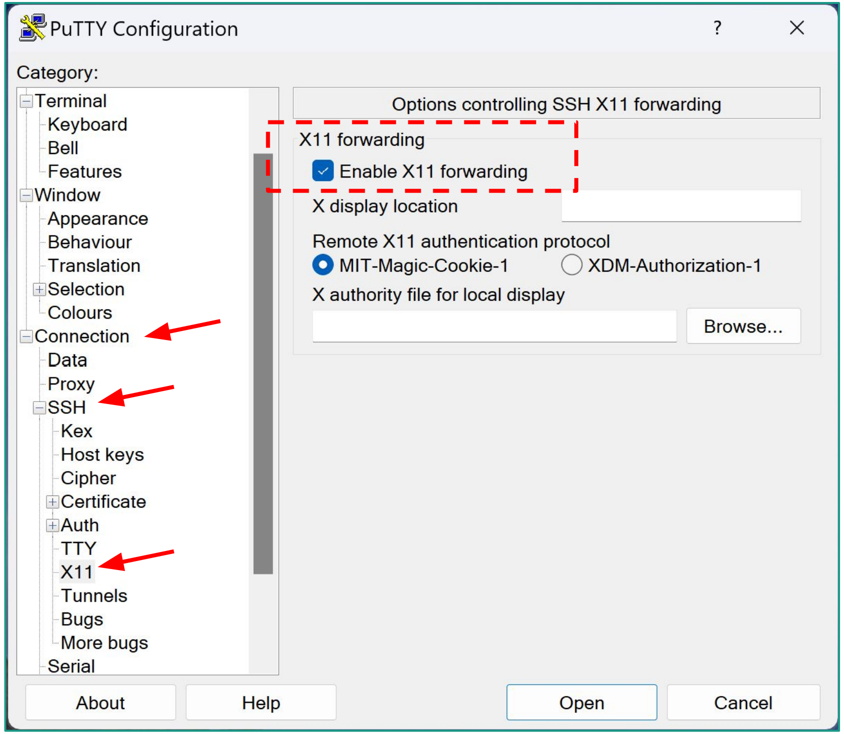
Download	Version	Released	Info
XQuartz-2.8.5.pkg	2.8.5	2023-01-26	For macOS 10.9 or later

Day1: Login to Hercules (Windows)

- Open the application **Putty**:

(optional) In case you want to use Xming





Note) Application for GUI on Windows: Xming (optional)

Day1: Login to Hercules (Windows: PowerShell)

```
(base) jongkim@RVLL-51106309:~$ ssh -X jongkim@hercules-login.hpc.msstate.edu
The authenticity of host 'hercules-login.hpc.msstate.edu (130.18.14.153)' can't be
established.
ED25519 key fingerprint is
SHA256:b***************************************************************.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? Yes

********** N O T I C E **********

(jongkim@hercules-login.hpc.msstate.edu) Password:
(jongkim@hercules-login.hpc.msstate.edu) Duo two-factor login for jongkim

Enter a passcode or select one of the following options:

1. Duo Push to XXX-XXX-5053

Passcode or option (1-1): 5**********
Success. Logging you in...
```

Day1: Basic HPC Information on Hercules

- Check your **user name** / **project account** / **Slurm QOS** / **work directory**:

```
(base) [jongkim@hercules-login-3 jongkim]$ whoami
jongkim
(base) [jongkim@hercules-login-3 jongkim]$ echo $USER
jongkim
(base) [jongkim@hercules-login-3 jongkim]$ module load
contrib noaatools
(base) [jongkim@hercules-login-3 jongkim]$ saccount_params

     Project: epic-explorer
          FairShare=1.000 (1/65)
          Partition Access: ALL
          Available QOSes:
batch,debug,novel,ood,urgent,windfall

          Directory: /work/noaa/epic-explorer DiskInUse=0 GB, Quota=2850 GB, Files=15,
FileQUota=0
          Directory: /work2/noaa/epic-explorer DiskInUse=19 GB, Quota=95000 GB, Files=1061,
FileQUota=0
```

Hercules QoS's and Limits

QoS	Priority	MaxNodes	MaxTime	Notes
batch	20	250	8 Hours	NOAA: Default QoS
novel	50	UNLIMITED	8 Hours	NOAA: Massive Jobs By request and approval only MinNodes=251
urgent	40	250	8 Hours	NOAA: High-Priority Jobs MaxJobsPerAccount=1
windfall	1	250	8 Hours	NOAA: Low-Priority Jobs
debug	30	250	30 Minutes	NOAA/MSU: Debugging Jobs MaxJobsPerUser=2
ood	1	1	8 Days	NOAA/MSU: Open OnDemand Jobs MaxCPUSPerNode=2
normal	20	5	2 Days	MSU: Default QoS
special	20	UNLIMITED	UNLIMITED	MSU: By request and approval only
priority	50	UNLIMITED	UNLIMITED	MSU: By request and approval only

- **Work space: /work2/noaa/epic-explorer/[username]**

Day1: Work Directory on Hercules

- Move to your **work2/noaa/epic-explorer** directory:

```
(base) [jongkim@hercules-login-3 ~]$ cd /work2/noaa/epic-explorer
(base) [jongkim@hercules-login-3 ~]$ mkdir -p $USER
(base) [jongkim@hercules-login-3 ~]$ cd $USER
(base) [jongkim@hercules-login-3 jongkim]$ git clone
https://github.com/NOAA-EPIC/CADRE-DA-training
Cloning into 'CADRE-DA-training'...
…………..
(base) [jongkim@hercules-login-3 jongkim]$ ls -R
.:
CADRE-DA-training

./CADRE-DA-training:
README.md  archive  diagnostic  year2_cases
```

Day1: Work Directory on Hercules

- Pre-run experiments are available at:

```
(base) [jongkim@hercules-login-3 cadre2026]$ cd
/work2/noaa/epic-explorer/cadre2026
(base) [jongkim@hercules-login-3 cadre2026]$ ls
cadre26.8895896.day1_ctrl                 cadre26.8900895.atms_err
cadre26.8895942.day2_hyb_weight           grid
cadre26.8896455.day2_nicas_length_scale   hercules.anaconda
cadre26.8896479.day3_atms_thinning        input_data
cadre26.8897035.day3_atms_err08           single_obs
cadre26.8900653.day3_no_atms
```

- Note: **`grid`** and **`input_data`** files are available as well as the anaconda modulefile that will be used in the hands-on experiment sessions.

Day1: Structure of JEDI Input YAML File (Top-level)

```
cost function:
  cost type: JEDI DA algorithm (3D-Var, 3D-FGAT, 4D-Var)
  time window: Assimilation time window
  geometry: Geometry of model (FV3)
  analysis variables: Variables used for analysis
  background: Background file (output of model forecast)
  background error: Background error covariance matrix (B-matrix): hybrid of
    SABER and ensemble, localization (SABER/BUMP_NICAS)
  observations: List of observation files (5 obs: conventional_ps,
    gnssro_cosmic2, ATMS N20, ascatw_ascat_metop-b, satwind_goes-16)
variational:
  minimizer: DRPCG (Derber-Rosati Preconditioned Conjugate Gradient
  iterations:
final:
  diagnostics:
  prints: print frequency
  increment: output increment
```

Day1: Experiment YAMLs Part 1

```yaml
cost function:
 cost type: 3D-Var
 jb evaluation: false
 time window:
   begin: '2024-02-23T21:00:00Z'
   length: PT6H
   bound to include: begin
 geometry:
   fms initialization:
     namelist filename:
./fv3jedi/fmsmpp.nml
     field table filename:
./fv3jedi/field_table
   akbk: ./fv3jedi/akbk.nc4
   layout:
   - 4
   - 4
   npx: 97
   npy: 97
   npz: 127
 analysis variables:
 - eastward_wind
 - northward_wind
 - air_temperature
 - ...
 background:
   datapath: ./bkg
   filetype: cube sphere history
   provider: ufs
   datetime: '2024-02-24T00:00:00Z'
   filenames:
   - ufsda.t18z.atm.f006.cubed_sphere_grid.nc
   - ufsda.t18z.sfc.f006.cubed_sphere_grid.nc
   max allowable geometry difference: 1e-4
   state variables: [...]
   field io names:
     eastward_wind: ugrd
     ...
 background error:
   covariance model: hybrid
   components:
   - covariance:
      covariance model: SABER
      saber central block:
        saber block name: gsi static covariance
        read:
          gsi akbk: ./fv3jedi/akbk.nc4
          gsi error covariance file:
./berror/gsi-coeffs-gfs-global.nc
          gsi berror namelist file:
./berror/gfs_gsi_global.nml
          processor layout x direction: 12
          processor layout y direction: 8
          debugging mode: false
          ...
    - covariance:
      covariance model: ensemble
      members from template:
        template:
          datetime: '2024-02-24T00:00:00Z'
          filetype: cube sphere history
          provider: ufs
          state variables: [...]
localization:
        localization method: SABER
        saber central block:
          saber block name: BUMP_NICAS
          active variables: [...]
          read:
            general:
              universe length-scale: 4641.0e3
            drivers:
              multivariate strategy: duplicated
              compute nicas: true
            model:
              level for 2d variables: last
            nicas:
              resolution: 6
              explicit length-scales: true
              horizontal length-scale:
              - groups: [...]
              vertical length-scale: [...]
      linear variable change:
        linear variable change name:
Control2Analysis
        input variables: [...]
        output variables: [...]
    weight:
      value: 0.875
```

Day1: Experiment YAMLs Part 2

```
 observations:
   obs perturbations: false
   observers:
   - obs space:
      name: conventional_ps
      obsdatain:
        engine:
          type: H5File
          obsfile:
./obs/obs.20240224.t00z.conventional_ps.nc
          missing file action: warn
      obsdataout:
        engine:
          type: H5File
          obsfile:
./output/diag.conventional_ps_2024022400.nc
      io pool:
        max pool size: 1
      simulated variables:
      - stationPressure
    obs operator:
      name: SfcCorrected
      variables:
      - name: stationPressure
      correction scheme to use: GSL
      station_altitude: height
      geovar_sfc_geomz:
height_above_mean_sea_level_at_surface
      geovar_geomz: geopotential_height
```

```
    linear obs operator:
      name: Identity
      variables:
      - name: stationPressure
    obs prior filters:
    - filter: Variable Assignment
      assignments:
      - name:
InputObsError/stationPressure
        type: float
        source variable:
ObsErrorData/stationPressure
       ...
variational:
 minimizer:
   algorithm: DRPCG
 iterations:
 - ninner: 2
   gradient norm reduction: 1e-10
   test: true
   geometry:
     fms initialization:
       namelist filename:
./fv3jedi/fmsmpp.nml
       field table filename:
./fv3jedi/field_table
     akbk: ./fv3jedi/akbk.nc4
     layout: ...
     vert coordinate: logp
   diagnostics:
     departures: bkgmob
```

```
 - ninner: 4
   gradient norm reduction: 1e-10
   test: true
   Geometry: ...
   diagnostics:
     departures: bkgmob1
final:
 diagnostics:
   departures: anlmob
 prints:
   frequency: PT3H
 increment:
   output:
     state component:
       filetype: cube sphere history
       filename:
./output/cubed_sphere_grid_atminc.jed
i.nc
       provider: ufs
       fields to write:
       - eastward_wind
       ...
       field io names:
         eastward_wind: ugrd
         ...
   geometry:
     ...
final j evaluation: false
```

Day1: Experiment Setup and Run

- Clone the GitHub **repository** for this tutorial
- Move into the **year2_cases** directory

```
cd  /work2/noaa/epic-explorer/$USER
git clone https://github.com/NOAA-EPIC/CADRE-DA-training.git
cd CADRE-DA-training/year2_cases
```

- **year2_cases** directory structure

```
(base) [jongkim@hercules-login-3 year2_cases]$ ls
build_bundle.sh  input_yaml              year2_aws_setup
build_gdas.sh    run_3dvar_hercules.sh
(base) [jongkim@hercules-login-3 year2_cases]$ cd input_yaml && ls
Day1  Day2  Day3
(base) [jongkim@hercules-login-3 year2_cases]$ cd Day1 && ls
jedi_3dvar_fv3_2024022400.yaml    jedi_3dvar_fv3inc_2024022400.yaml
```

Note: jedi_3dvar_fv3*.yaml (FV3-JEDI input yaml)
jedi_3dvar_fv3inc*yaml (GDASapp UFS increment handling yaml)

Day1: Slurm Job Submission Script

- Move to the CADRE-DA-training/year2_cases directory and open the job card:

```
(base) [jongkim@hercules-login-3 year2_cases]$ cd
/work2/noaa/epic-explorer/jongkim/CADRE-DA-training/year2_cases
(base) [jongkim@hercules-login-3 year2_cases]$ ls
build_bundle.sh  build_gdas.shinput_yaml  run_3dvar_hercules.sh  year2_aws_setup
(base) [jongkim@hercules-login-3 year2_cases]$ more run_3dvar_hercules.sh
#!/bin/bash -l
#SBATCH --account=epic
#SBATCH --job-name=fv3jedi
#SBATCH --output=log.cadre26.%j
#SBATCH --partition=hercules
#SBATCH --qos=batch
#SBATCH --time=00:20:00
#SBATCH --nodes=2
#SBATCH --ntasks-per-node=48
#SBATCH --mem=128G
###SBATCH --exclusive
set -xue
ulimit -s unlimited; ulimit -a;
# Parameters
# Path to JEDI bin directory: change this if you built this in another directory
JEDI_BIN_PATH="/work2/noaa/epic/chjeon/cadre26_hercules/GDASApp/build/bin"
# Path to input files (pre-staged)
JEDI_INPUT_PATH="/work2/noaa/epic-explorer/cadre2026/input_data"
```

Day1: Executables: GDASapp, FV3 JEDI Bundle

```
(base) [jongkim@hercules-login-4 bin]$ cd
/work2/noaa/epic/chjeon/cadre26_hercules/GDASApp/build/bin
(base) [jongkim@hercules-login-4 bin]$ ls *.x
fv3jedi_letkf.x
fv3jedi_plot_field.x
fv3jedi_processperts.x
fv3jedi_tlm_toolbox.x
fv3jedi_var.x
gdas.x
gdas_fv3jedi_add_increment.x
gdas_fv3jedi_chem_diagb.x
gdas_fv3jedi_correction_increment.x
gdas_fv3jedi_ensemble_add_increment.x
gdas_fv3jedi_error_covariance_toolbox.x
gdas_fv3jedi_fv3inc.x
gdas_fv3jedi_land_ensrecenter.x
gdas_fv3jedi_scf_to_ioda.x
gdas_ioda-stats.x
```

Day1: Experiment Setup and Run

- Copy the Day 1 JEDI YAMLs into the input_yaml directory and submit the Slurm sbatch job card:

```
cp ./input_yaml/Day1/*.yaml ./input_yaml
sbatch run_3dvar_hercules.sh
Submitted batch job 8908314
```

- Monitor the status of the submitted job and the exp_case output directory:

```
(base) [jongkim@hercules-login-3 year2_cases]$ squeue -u $USER
        JOBID PARTITION     NAME     USER   ST    TIME  NODES NODELIST(REASON)
      8908314  hercules  fv3jedi  jongkim    R    0:44      2 hercules-04-[49-50]
(base) [jongkim@hercules-login-3 year2_cases]$ cd exp_case/cadre26.8908314/
```

Day1: Experimental Output Directory

```
(base) [jongkim@hercules-login-4 exp_case]$ tree cadre26.8908314
cadre26.8908314
├── OUTPUT.fv3inc
├── OUTPUT.fv3jedi
├── berror -> /work2/noaa/epic/chjeon/cadre26/input_data/berror
├── bkg -> /work2/noaa/epic/chjeon/cadre26/input_data/bkg
├── crtm -> /work2/noaa/epic/chjeon/cadre26/input_data/crtm
├── ens -> /work2/noaa/epic/chjeon/cadre26/input_data/ens
├── errfile_fv3jedi
├── errfile_inc
├── fv3jedi -> /work2/noaa/epic/chjeon/cadre26/input_data/fv3jedi
├── grid -> /work2/noaa/epic/chjeon/cadre26/input_data/grid
├── jedi_3dvar_fv3_2024022400.yaml
├── jedi_3dvar_fv3inc_2024022400.yaml
├── logfile.000000.out
├── obs -> /work2/noaa/epic/chjeon/cadre26/input_data/obs
├── output
│   ├── cubed_sphere_grid_atminc.jedi.nc
│   ├── diag.atms_n20_2024022400.nc
│   ├── diag.conventional_ps_2024022400.nc
│   ├── diag.gnssro_cosmic2_2024022400.nc
│   ├── diag.satwnd.abi_goes-16_2024022400.nc
│   ├── diag.scatwnd.ascat_metop-b_2024022400.nc
│   ├── obs.20240224.t00z.atms_n20.satbias.nc
│   ├── obs.20240224.t00z.atms_n20.satbias_cov.nc
│   ├── ufsda.t00z.atminc.cubed_sphere_grid.tile1.nc
│   ├── ufsda.t00z.atminc.cubed_sphere_grid.tile2.nc
│   ├── ufsda.t00z.atminc.cubed_sphere_grid.tile3.nc
│   ├── ufsda.t00z.atminc.cubed_sphere_grid.tile4.nc
│   ├── ufsda.t00z.atminc.cubed_sphere_grid.tile5.nc
│   └── ufsda.t00z.atminc.cubed_sphere_grid.tile6.nc
├── output_ref -> /work2/noaa/epic/chjeon/cadre26/input_data/output_ref
├── py_ne -> /work2/noaa/epic/chjeon/cadre26/input_data/py_ne
└── warnfile.000000.out
```

Completed experimental case directory

JEDI input directories and yaml files

output directory of FV3-JEDI

JEDI increment file

IODA H(x) outputs for observations

Tiled UFS increment files

Day1: Experiment Output Files for Analysis Diagnostics

- Move to the **output** directory:

```
(base) [jongkim@hercules-login-3 year2_cases]$ cd output
(base) [jongkim@hercules-login-3 year2_cases]$ ls
```

JEDI increment file
(output of ‘fv3jedi_var.x’)

JEDI H(x) analysis
(output of ‘fv3jedi_var.x’)

Increment files for FV3 (6tiles)
(output of ‘gdas_fv3jedi_fv3inc.x’)

```
cubed_sphere_grid_atminc.jedi.nc
diag.atms_n20_2024022400.nc
diag.conventional_ps_2024022400.nc
diag.gnssro_cosmic2_2024022400.nc
diag.satwnd.abi_goes-16_2024022400.nc
diag.scatwnd.ascat_metop-b_2024022400.nc
obs.20240224.t00z.atms_n20.satbias.nc
obs.20240224.t00z.atms_n20.satbias_cov.nc
ufsda.t00z.atminc.cubed_sphere_grid.tile1.nc
ufsda.t00z.atminc.cubed_sphere_grid.tile2.nc
ufsda.t00z.atminc.cubed_sphere_grid.tile3.nc
ufsda.t00z.atminc.cubed_sphere_grid.tile4.nc
ufsda.t00z.atminc.cubed_sphere_grid.tile5.nc
ufsda.t00z.atminc.cubed_sphere_grid.tile6.nc
```

Input of ATMS for next cycle (not used here)

Day1: Experiment Output: Diagnostics

- Once the FV3-JEDI jobs finish, the UFS-DA Diagnostics toolkit can be applied to the output files. UFS-DA Diagnostics package was pre-installed on Hercules. Example diagnostic yams are available.

```
source /work2/noaa/epic-explorer/cadre2026/hercules.anaconda
cd  /work2/noaa/epic-explorer/$USER/CADRE-DA-training/diagnostic
(base) [jongkim@hercules-login-4 diagnostic]$ ls yamls/day1/
increment_maps.yaml  obs_diag.yaml  spectra_bkg_inc.yaml
```

- Once users update example diagnostic yaml files, diagnostic tools can be run:

```
ufsda-spectra-bkg-inc --yaml path/to/day1/spectra_day1.yaml
ufsda-inc-maps --yaml path/to/day1/increment_maps_day1.yaml
ufsda-obs-diag --yaml path/to/day1/obs_diag_day1.yaml
ufsda-jedi-log /work2/noaa/epic-explorer/CADRE2026/cadre26.8434573.day1/OUTPUT.fv3jedi \
    --output day1_log_report.txt
```

- Note that runtime arguments are used for jedi log parser: input jedi log file and output file name.

Day1: Experiment Output: Diagnostics

- Key diagnostic yaml parameters that users need to update:
  - **increment_maps.yaml:** vars, levels, zonal mean plot triggers, experiments name and prefix, grid file prefix (note that users can use /work2/noaa/epic/CADRE2026/grid/C96_grid.tile*), output_dir, color scale amplification
    - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/diagnostic/yamls/day1/increment_maps.yaml
  - **obs_diag.yaml:** output_dir, prefix_root, list of observations with diagnostic option triggers, atms omb scatter plots on map
    - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/diagnostic/yamls/day1/obs_diag.yaml
  - **spectra_bkg_inc.yaml**: vars, background atm_file, increments prefix, grid file and vars, spectra vars and output_dir
    - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/diagnostic/yamls/day1/spectra_bkg_inc.yaml

Day1: Experiment Output: Diagnostics

- Use SCP to copy the generated PNG files from the HPC system to your local machine, regardless of whether you're on macOS, Windows PowerShell, or Windows WSL.:

```
scp <username>@hercules-login.hpc.msstate.edu:/path/to/*.png .
or
scp -r <username>@hercules-login.hpc.msstate.edu:/path/to/png_directory .
```

Day1: Control Experiment: Key Diagnostics Part 1

- **Increment Maps:** Look for physically meaningful spatial patterns; noisy patches may indicate QC or background-error issues.
- **Zonal-Mean Increments:** Check vertical structure and hemispheric balance; watch for unrealistic spikes.
- **Obs-Space Histograms:** Should be roughly centered and bell-shaped; shifts indicate bias, wide tails indicate QC or error-model problems.
- **Extended RMS Stats:** Quick indicator of fit to observations; compare OMB vs OMA to confirm improvement.

Day1: Control Experiment: Key Diagnostics Part 2

- **Scan-Position Checks:** Identify cross-track striping or limb-angle artifacts (especially for radiances).
- **Latitude-Binned Stats:** Look for hemispheric regional imbalance in observation fits.
- **Spectral Analysis:** BKG vs INC to show how increments distribute across scales.
    - Large-scale dominance → balanced, physically meaningful corrections
    - Excess small-scale energy → noise, sampling issues, or mis-specified B
    - Compare BKG and INC curves to see where DA is adding information
- **OMA Jo/p:** a heuristic sanity check for formal chi-square consistency theory.
    - Does the analysis fit the observations better than the background?

Day1: (Pre-run) Diagnostics — Overview

- Diagnostic output examples: here
- **inc_plots** — Horizontal increment maps for key variables.
  - Look for physically meaningful structures, smoothness, and absence of noisy small-scale artifacts.
- **obs-diag** — Observation-space diagnostics: histograms, OMB/OMA, RMS, scan-angle checks, latitude bins.
  - Focus on bias signatures, tail behavior, and whether OMA improves over OMB.
- **spectra-bkg-inc** — Spectral energy comparison between background and increments.
  - Look for expected slopes (planetary, synoptic, mesoscale) and check for excess small-scale increment energy.
- **jedi-log** — JEDI variational summary, including OMA Jo/p and configuration metadata.
  - Use Jo/p only as a heuristic sanity check.

Day1: Optional Hands-On: Single-Observation Experiment

- Single observation extraction tool is provided at https://ufs-da-diagnostics.readthedocs.io/en/latest/usage_single_obs.html
  - This additional experiment is useful for debugging, training, and controlled experiments such as single-observation impact studies.
- A ready-to-run ctrl example is provided at https://github.com/NOAA-EPIC/CADRE-DA-training/tree/main/year2_cases/input_yaml/single_obs/ctrl
  - ATMS single-observation file (one footprint randomly selected in the East Pacific tropical region)
- These small experiments are useful because they let you:
  - See how a single observation influences the analysis
  - Build intuition for increments, OMB/OMA, and error settings
  - Prepare for deeper intuitive diagnostics on Day-2 and Day-3
- Even one quick run helps make the DA system’s behavior much more concrete.

Day 2: SABER/BUMP Background Error Experiments

- By the end of this session, you will be able to:
  - Understand the role of SABER/BUMP in modeling background error covariance
  - Modify NICAS correlation-length parameters in YAML and observe their effect on increment structure
  - Adjust hybrid weights (static vs. ensemble) and compare analysis increments
  - Interpret power spectra and increment spread diagnostics
  - Use the diagnostic tool to generate cross-experiment comparison panels
  - Understand how NICAS horizontal length-scale (in meters, specified as Gaspari-Cohn support radius) controls the spatial smoothness of the static B correlation — e.g., increasing from 500 km to 600 km broadens increments

Day2: Background-Error Covariance Model Experiments

Experiment	What changes in Yaml
Exp 2 — Horizontal Length Scale	Modify `horizontal-length-scale` in the `BUMP_NICAS` YAML block (e.g., `500e+3 → 1000e+3 m`); compare increment spread, power spectra, and effective resolution
Exp 3 — Hybrid Weight	`static_weight` / `ensemble_weight` ratio

- Day2 experiments focus on how the B-model settings shape increments, spectra, and overall analysis behavior.
- Run the single-obs experiment (if time allows) for a clearer view of B-model effects.

Interactive Session: Day 2

https://ufs-da-diagnostics.readthedocs.io/en/latest/cadre2026_epic.html

Day2: Operators of Background Error Covariance

- **Background error covariances — static, ensemble, and hybrid — are all built from a common set of decomposed operators:**

$$B = T\ L\ D\ L^T\ T^T$$

  - **Balance operator:** Shapes physically balanced increments (mass–wind–temperature relationships)
  - **Correlation operator:** Spreads increments horizontally and vertically
  - **Variance operator:** Sets the magnitude of the increments
- Day1 Control experiment: Hybrid B is constructed from two separate background-error components
  - A static SABER block (gsi static covariance) and an ensemble block (covariance model: ensemble).
  - $B_{hybrid}=\alpha B_{static}+(1-\alpha)B_{ens}$: control experiment (α=0.125)
    - Static B = stability and smoothness
    - Ensemble B = flow-dependent structure

Day2: B-Model Sensitivity – NICAS Length Scale

- Exp 2 modifies the L operator (NICAS horizontal correlation length scale) in the ensemble-covariance part of the hybrid B.
  - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/year2_cases/input_yaml/Day2/exp_hyb_weight/jedi_3dvar_fv3_2024022400.yaml

```
##################################################
# CADRE 2026, Day2, exp_nicas_length_scale
##################################################
                horizontal length-scale:
                - groups:
                  - common
                  profile:
                    - 5569200.0
                    - 5569200.0
                    - 5569200.0
                    - 5569200.0
                    - 5569200.0
………. .
                    - 1499400.0
                    - 1499400.0
                    - 1499400.0
```

Day2: B-Model Sensitivity: Hybrid Weight (Static/Ens Ratio)

- Exp 3 adjusts the hybrid static/ensemble weight, altering the relative influence of ensemble vs static covariance.
  - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/year2_cases/input_yaml/Day2/exp_hyb_weight/jedi_3dvar_fv3_2024022400.yaml

```
#################################################
# CADRE 2026, Day2, exp_hyb_weight
#################################################
      weight:
        value: 0.6
#################################################
```

```
#################################################
# CADRE 2026, Day2, exp_hyb_weight
#################################################
      weight:
        value: 0.4
#################################################
```

Day2: Experiment Setup and Run

- Copy the Day 2 JEDI YAMLs into the input_yaml directory and submit the Slurm sbatch job card:

```
cp ./input_yaml/Day2/exp_hyb_weight/*.yaml ./input_yaml
sbatch run_3dvar_hercules.sh
Submitted batch job 8908314


cp ./input_yaml/Day2/exp_nicas_length_scale/*.yaml ./input_yaml
sbatch run_3dvar_hercules.sh
Submitted batch job 8908314
```

- Once the FV3-JEDI jobs finish, the UFS-DA Diagnostics toolkit can be applied to the output files

```
ufsda-spectra-bkg-inc --yaml path/to/day2/spectra_day2.yaml
ufsda-inc-maps --yaml path/to/day2/increment_maps_day2.yaml
ufsda-obs-diag --yaml path/to/day2/obs_diag_day2.yaml
ufsda-jedi-log /work2/noaa/epic-explorer/cadre2026/cadre26.day2.hyb_weight/OUTPUT.fv3jedi \
    --output jedi_log.out
```

Day2: Sensitivity Diagnostics (Static Weight ↑, NICAS Scale ↑)

- **Hybrid Static Weight ↑:**
  - Increments become larger-scale, smoother; mesoscale energy drops; static B dominates structure.
- **NICAS Length-Scale ↑:**
  - Broader spatial influence; wider, smoother increments; spectra shift toward planetary/synoptic scales.
- **Increment Maps:**
  - CTRL: sharper; Static↑: smoother; NICAS↑: broadest.
  - Focus on how spatial structure changes, not magnitude.
- **Spectral Comparison:**
  - Static↑: mesoscale suppression; NICAS↑: enhanced large-scale dominance.
  - Participants may identify slope shifts.
- **Obs-Space Stats:** Expect small OMB/OMA changes; these tests affect structure, not fit.

Day2: (Pre-computed) Diagnostics: Overview & Discussion

- Pre-run diagnostic outputs for Exp2 and Exp3:
  - https://github.com/jkbk2004/CADRE-DA-training/tree/diag-results/diagnostic/results/year2_cases/day2_nicas_length_scale
  - https://github.com/jkbk2004/CADRE-DA-training/tree/diag-results/diagnostic/results/year2_cases/day2_hyb_weight
- inc_plots — Horizontal increment maps for key variables.
  - Participants should look for how static-weight↑ and NICAS-scale↑ change smoothness, spatial spread, and mesoscale suppression.
- Observation-space diagnostics: histograms, OMB/OMA, RMS, scan-angle checks, latitude bins.
  - Focus on bias signatures, QC impacts, and whether OMA improves over OMB after bias/QC adjustments.
- spectra-bkg-inc — Spectral energy comparison between background and increments.
  - Look for expected slopes (planetary, synoptic, mesoscale) and how static-weight↑ or NICAS-scale↑ shift energy toward larger scales.
- jedi-log — JEDI variational summary, including OMA Jo/p and configuration metadata.
- Worth to run quick single obs cases:
  - https://github.com/jkbk2004/CADRE-DA-training/tree/diag-results/year2_cases/input_yaml/single_obs

Day 3: Observation Thinning & Error Experiments

- By the end of this session, you will be able to:
  - Modify ATMS spatial thinning parameters in YAML and assess observation density changes
  - Scale ATMS observation errors and interpret OMB/OMA distribution shifts
  - Toggle observation types on/off and evaluate their impact on the analysis
  - Generate observation density maps, OMB/OMA histograms, and QC flag summaries
  - Compare all observation experiments against the Day-1 control baseline

Experiment	What Changes in the YAML?
Exp 4 — ATMS Thinning	`thinning_mesh` parameter (60 km → 120 km) Reduces observation density by increasing the spatial thinning distance. Fewer observations assimilated per cycle.
Exp 5 — ATMS Obs Error	`obs_error_scale` multiplier (1.5×, 2.0×) Inflates prescribed observation errors, reducing the weight of observations relative to the background.
Exp 6 — Obs Toggle (Optional)	Enable/disable ATMS, GNSSRO or surface pressure Assesses the individual contribution of specific observation types to the analysis quality.

Interactive Session: Day 3

https://ufs-da-diagnostics.readthedocs.io/en/latest/cadre2026_epic.html

Day 3: ATMS Thinning, Error Inflation, and No-ATMS Tests

- **Experiment Scopes:**
  - **Exp 4 — ATMS Gaussian Thinning**
    - Apply Gaussian thinning to reduce dense ATMS footprint clusters.
    - Shows how smoother spatial sampling affects QC acceptance, radiance usage, and increment smoothness.
  - **Exp 5 — ATMS Error Inflation**
    - Increase (or channel-tune) ATMS observation errors to weaken radiance influence.
    - Examine changes in Jo/p, increment amplitude, and vertical temperature structure.
  - **Exp 6 — No-ATMS Experiment**
    - Remove all ATMS radiances to isolate their net contribution.
    - Compare increments, spectra, and upper-tropospheric temperature fields against the control.

Day 3: ATMS Gaussian Thinning (YAML Block)

- Apply Gaussian thinning to reduce dense ATMS footprint clusters and produce smoother, more uniform radiance sampling
  - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/year2_cases/input_yaml/Day3/exp_atm_thinning/jedi_3dvar_fv3_2024022400.yaml

```
################################################
# CADRE 2026, Day3, exp_gaussian_thinning
################################################
      - filter: Gaussian Thinning
        horizontal_mesh: 300
################################################
```

Day 3: ATMS Gaussian Thinning (YAML Block)

- Apply Gaussian thinning to reduce dense ATMS footprint clusters and produce smoother, more uniform radiance sampling
  - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/year2_cases/input_yaml/Day3/exp_atm_thinning/jedi_3dvar_fv3_2024022400.yaml

```
#################################################
# CADRE 2026, Day3, exp_gaussian_thinning
#################################################
      - filter: Gaussian Thinning
        horizontal_mesh: 300
#################################################
```

Day 3: ATMS Observation Error Inflation (YAML Block)

- Adjust ATMS observation errors (uniform or channel-dependent) to control how strongly ATMS radiances influence the analysis.

```
##################################################
# CADRE 2026, Day3, exp_obs_error
##################################################
      #- filter: Perform Action
      #  filter variables:
      #  - name: brightnessTemperature
      #    channels: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22
      #  action:
      #    name: inflate error
      #    inflation factor: 0.8
####################
      # Window channels (1–5)
      - filter: Perform Action
        filter variables:
        - name: brightnessTemperature
          channels: 1-5
        action:
          name: inflate error
          inflation factor: 0.55
```

Day 3: Experiment with No ATMS

- Remove all ATMS radiances to isolate their net impact on the analysis and understand how the system behaves without microwave temperature sounders.
    - https://github.com/NOAA-EPIC/CADRE-DA-training/blob/main/year2_cases/input_yaml/Day3/exp_no_atms/jedi_3dvar_fv3_2024022400.yaml
- What to look for:
    - Differences in increment patterns compared to the control experiment
    - Reduced upper-tropospheric temperature constraint
    - Changes in spectra and vertical structure due to missing microwave temperature information
    - Clear visualization of ATMS' contribution when comparing against the control experiment side-by-side

Day 3: Key Diagnostics to Focus On (ATMS Sensitivity)

- Pre-run diagnostic outputs for Exp4/5/6: https://github.com/jkbk2004/CADRE-DA-training/tree/diag-results/diagnostic/results/year2_cases
  - Folders: day3_atms_thinning, day3_atms_err, day3_no_atms
- ATMS Thinning:
  - fewer obs → smaller increments, weaker vertical structure, reduced mesoscale energy.
- Uniform Error Reduction (×0.8):
  - Smaller R → larger increments, stronger OMA improvement, Jo/p increases due to R^{-1} scaling.

Day 3: Key Diagnostics to Focus On (ATMS Sensitivity)

- Channel-Dependent Error Tuning:
  - Channels with reduced error → stronger increments; inflated error → weaker increments.
  - Expect Jo/p to increase for channels where R is reduced.
- No-ATMS Experiment:
  - Much weaker temperature-sensitive increments; reduced vertical coupling; smoother spectra.
  - Jo/p increases because the system under-fits without ATMS.
- Spectral Comparison:
  - Thinning → mesoscale drop
  - Error↓ → mesoscale rise
  - No-ATMS → overall energy reduction

Day 3: Jo/p and RMS□^{2} Interpretation

- OMA Jo/p: quantitatively meaningful messages
    - Error↓ → Jo/p ↑ (because R^{-1} increases faster than OMA decreases)
    - Thinning → Jo/p ↑ (under-fitting)
    - No-ATMS → Jo/p ↑ strongly
- RMS□^{2}:
    - RMS□^{2} should track the same trend as Jo/p:
    - Error↓ → RMS□^{2} ↑
    - Thinning → RMS□^{2} ↑
    - No-ATMS → RMS□^{2} ↑ significantly
- Consistency Check: Participants may verify
    - Jo/p ↑ ↔ RMS□^{2} ↑ ↔ increment magnitude changes
    - across all ATMS sensitivity experiments.
- Key Message: Day 3 experiments show how DA responds quantitatively to changes in R and observation density.

Take-Home Messages: Days 1-3

- **You control the DA system** — understanding the workflow, directories, YAML structure, and experiment setup is essential for reproducible science.
- **Background error modeling matters** — NICAS length scales and hybrid static/ensemble weights directly shape increment patterns, flow-dependence, and effective resolution.
- **Observations are not equal** — thinning, error tuning, and sensor removal (ATMS on/off) all change Jo/p, QC behavior, and the vertical/ horizontal structure of increments.
- **Small YAML edits → big scientific differences** — each experiment (Exp 1–6) demonstrates how targeted configuration changes reveal system behavior and observation impact.
- **Diagnostics are your compass** — increments, spectra, OMB/OMA, Jo/p, and QC counts tell the story of what the DA system is doing and *why*.
- **Reproducibility + clarity = community strength** — clean experiment structure, clear documentation, and modular YAMLs make it easy for others to run, compare, and extend your work.

Thanks!
Request for EPIC Support

- emails:
  - support.epic@noaa.gov or jong.kim@noaa.gov
- Git discussions:
  - https://github.com/NOAA-EPIC/CADRE-DA-training